

GAURAB ACHARYA

Vancouver, BC, Canada

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EDUCATION

University of British Columbia

Bachelor of Applied Science in Biomedical Engineering — Specialized in Systems and Signals

Sep. 2019 – Dec. 2024

Vancouver, British Columbia

TECHNICAL SKILLS

Languages: Java, Python, C++, C, SQL, Typescript, JavaScript, HTML, CSS, R, MATLAB, Swift

Technologies: Apache Airflow, Docker, Jenkins, Oracle, PostgreSQL, Clickhouse, Grafana

Frameworks: React, Spring Boot, TensorFlow, JUnit, R Shiny, Flask

EXPERIENCE

QuoteMedia

May 2024 – Aug. 2024, May 2022 – Apr. 2023

Software Engineer Intern

Vancouver, BC

- Improved database query throughput by 10x by tuning data schema and migrating legacy database from **Oracle** to **PostgreSQL**
- Contributed to the development and testing of a new data pipeline using **SQL**, **Python**, and **Apache Airflow** to migrate data from **Oracle** database to **Clickhouse** to 20x query efficiency on large aggregations for reports
- Created new API endpoints, from end to end, using **Java**, **SQL**, and **Spring Boot** to provide new financial data for clients
- Modernized financial API endpoints to improve data loading and response efficiency for a smooth financial chart experience
- Created an http load test pipeline using **Vegeta**, **Python**, and **Apache Airflow** to run and store results in **InfluxDB**, resulting in long term load test tracking in a dashboard setting
- Created dashboards in **Kibana** and **Grafana** to visualize key business metrics such as client API usage, speed, and load test results

ProCogia

May 2023 – Sep. 2023

R Developer Intern

Vancouver, BC

- Designed and created a R package, to analyze **R** scripts in less than 5 seconds, providing debug solutions, data operation tracking, and code assistance using the **OpenAI ChatGPT API**
- Constructed the package, using prompt engineering by setting AI system roles, and providing examples question and answer pairs, to generate results with 90% consistency
- Designed chat sessions to store results in memory, to increase the context window and improve response precision and accuracy

PROJECTS

UBC Capstone Medical Augmented Reality Device | *Unity, C#, ARKit*

- Developed an IOS app using **Unity** and **OpenXR** to render vein images for a Medical Device Startup, Vantech Medical, using **Apple ARKit's** image tracking and projection
- Hosted the client's Infrared (IR) Vein Detection Imaging Model on **Google Cloud** and integrated it with the Unity application, to capture the patient's arm and render the IR vein images in the same location in augmented reality with less than 1 second latency
- Conducted **UI testing** with healthcare workers, gathering user feedback to improve functionality and user experience

Posture Position Corrector | *UBC Biomedical Engineering Student Design Team* | *React Native, JavaScript*

- Designed an IOS and Android app, connected to an accelerometer and gyroscope, to support posture correction physiotherapy
- Developed posture position algorithms to send a notification when detecting 5° changes in posture.
- Built the app in **React Native** to connect with the hardware, to allow users to visualize the data and provide notifications

Sneaker Image Recognition App | *Hack the North Winner* | *Python, TensorFlow Lite, Swift*

- Trained an image classification machine learning model using **Python** and **TensorFlow Lite**, to recognize Jordan sneaker models
- Built an iOS application, using **Swift**, allowing users to upload photos to recognize the model of Jordan sneakers
- Contained the machine learning model on device using the **HOT-G API**, to ensure data security and speed

LEADERSHIP / EXTRACURRICULAR

Basketball / Coaching

Apr. 2016 – Present

RBL Basketball , *Drive Basketball* , *Strathcona Community Centre*

- 4x British Columbia Provincial Basketball Championship All-Star
- Coached over 1500 players in basketball camps and teams, working with three different clubs in Vancouver